

PEOPLE MOVEMENT SYSTEM

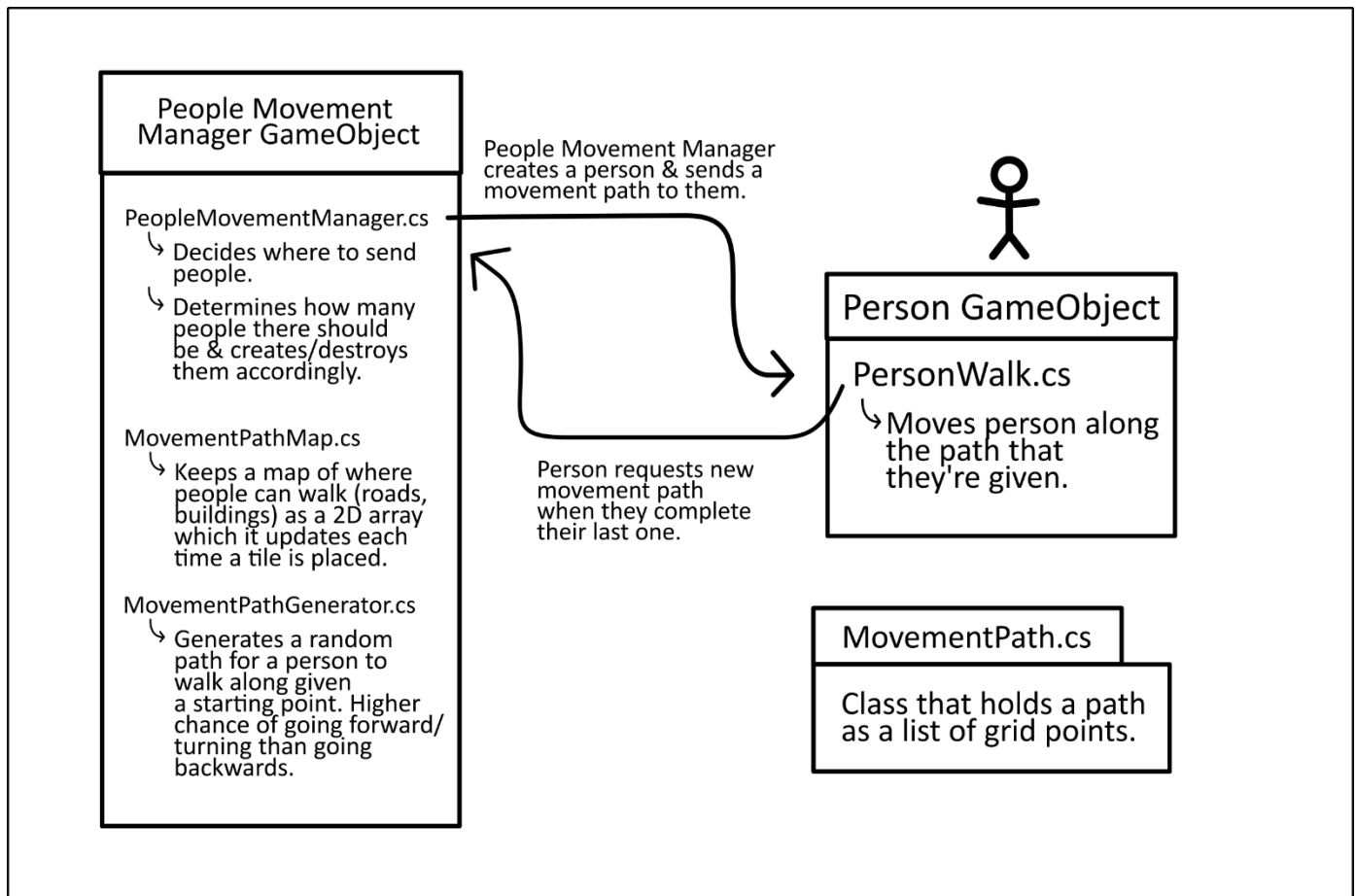
OVERVIEW

The People Movement System controls the tiny people that you can watch wander around on the roads. The primary scripts involved in this system are `PeopleMovementManager.cs`, `MovementPathMap.cs`, `MovementPathGenerator.cs`, and `PersonWalk.cs`. Note that this system doesn't deal with counting how many people there are or if there are enough employees for factories – all of that's handled by the People Manager. For more information on that, see the doc “`People_System.pdf`”.

There are a few ways that people can be created. The first is when the people button is pressed. In that case, a temporary person is created, and they directly walk into a random building. If the manager determines that it already has enough people, it will destroy that person once they get to the building. Otherwise, it will teleport that person to a random house and have them continue to wander around. The second way that people can be created is if the manager determines that there aren't enough people wandering around; in that case it creates a person in a random house and has them start wandering around, teleporting them to a new house each time they walk into a building. The final way people are created is when the player switches areas. When that happens, all of the people in the previous area are violently destroyed by the people movement manager, and then the correct number of people are created in random locations on the next area.

Note: It determines how many people there should be using this formula if there's more than 7 buildings on the current chunk: $\text{numOfPeople} = 7 + ((\text{numberOfBuildings} - 7) / 3)$. If there's less than 7 buildings, it's just $\text{numOfPeople} = \text{numberOfBuildings}$.

HOW THE SCRIPTS INTERACT



Note about movement paths: movement paths can either be direct paths, or road paths. Road paths start at a point and randomly move around along roads. They're used when the person is randomly wandering around. Direct paths are a path between two points. They're used when the person needs to be sent to a location, like right after they're placed when they need to be sent to a random building.

Note about a possible improvement to this system: It's kind of unnecessary to have people go through people movement manager to get a new path. They could really go directly to the movement path generator themselves.

INTERNAL API – PEOPLEMOVEMENTMANAGER.CS

Below are a few of the most important public functions of `PeopleMovementManager.cs`, which is accessible with the singleton `PeopleMovementManager.current`.

`void CreatePersonAtRandomLocation(bool instantiateInHouse)`

Creates a person at a random location. If the parameter is true, the person will be instantiated in a house; otherwise they will be instantiated on a road.

`NewPersonAdded()`

This function is called by the people button whenever it's pressed. It creates a new person and sends them to a random activated building.

`void PersonFinishedPath(PersonWalk person)`

This is called by a person whenever they finish a path. When that happens, the people movement manager either gives them a new path, teleports them to a new building if they're currently in a building, or destroys them in a fit of rage if it's determined that there are too many people.

`private void SendPersonToActivatedBuilding(PersonWalk person)`

This is a private function, but it seemed helpful to explain. Sends "person" to a random activated building using a direct movement path. It's used whenever a person is first created.

`private void SendPersonToActivatedRoad(PersonWalk person)`

This is also a private function, but it seemed helpful to explain. Sends "person" to a random activated road. This function is called by a person on themselves when they go into panic mode after they reach the end of their current path and realize they are no longer on a road (if, for example, the player deletes the road they were on). This is their last attempt to find a path they can walk along. If they can't find an activated road, they're destroyed.

INTERNAL API – OTHER SCRIPTS

MovementPathGenerator.cs

MovementPath MakeRandomPathAlongRoads(Vector3 worldStartLocation)

Returns a movement path that randomly travels along activated roads, starting at “worldStartLocation.” It automatically centers the path on the correct side of the road depending on the current direction of travel.

MovementPath MakeDirectPath(Vector3 worldStartLocation, Vector3 worldEndLocation, bool centerEndPathOnCell)

Returns a movement path that travels from “worldStartLocation” to “worldEndLocation”. The movement path it generates returns points that are set between grid cells, not centered on them. If centerEndPathOnCell is true, it will add a final path point that is centered on the grid cell at the end location.

MovementPathMap.cs

MapTileType GetTileForPoint(Vector3 tileWorldCoordinates)

Returns what type of tile is at “tileWorldCoordinates” using the MapTileType enum. Possible states of that enum are Empty, Road, Building. People are allowed to walk along Roads and Buildings. However, if they hit a building, the movement path must be ended there.

PersonWalk.cs

bool PersonIsOnActivatableBuilding()

Returns true or false depending on whether the given person is currently on a building. This is used to determine if a person ended their path on a building. If they did, the people movement manager can determine whether to teleport them to a new building or destroy them if there are too many people.

void RunAlongPath(MovementPath path)

This tells a person to begin moving along the provided “path.” Called by the people movement manager to send the person along a new path when they finish the old one.

void TeleportPersonToLocation(Vector3 location)

Teleports person to “location”. Used by the person movement manager when it teleports people to a new building.

-Graydon Bruyn, May 2026